



Movie Ticket Booking Management Application

National Internship Program (NIP) · Pega Platform™ · Pega Academy · 2026

Scenario

CineWave Entertainment manages movie ticket bookings across multiple theatres and locations. Ticket booking and tracking processes are currently handled manually through emails and offline systems, leading to delays, lack of visibility, and inefficiencies in managing bookings and customer communication.

CineWave Entertainment wants to build a **Movie Ticket Booking Management** application on Pega Platform™ to:

- Allow customers to request movie ticket bookings
- Enable staff to manage show details and seating availability
- Capture customer confirmations before final booking
- Process booking requests and track status
- Notify customers upon booking confirmation

Project Objectives

Students are expected to:

- Design a case lifecycle for ticket booking
- Define and implement user interactions
- Model required data structures
- Configure basic business logic and automation
- Ensure smooth flow from booking request to ticket confirmation

Prerequisite

Before starting this project, complete the **Pega Environment Setup and Application Creation Guide**. Use Pega Blueprint to generate your application scaffold, then complete the user stories below in your Academy instance.

User Stories

US-001: Submit Movie Ticket Request

User Story

A customer should be able to submit a movie ticket request in order to initiate the booking process.

Implementation Details

A case type named **Movie Ticket Request** should be created to represent the end-to-end booking process. Within this case type, an initial stage should be defined to capture booking request details from the customer. The **Customer** persona should be able to provide information such as **Movie Name**, **Show Date**, **Show Time**, and **Number of Tickets**. These inputs should be validated to ensure accuracy and completeness before submission. The case should also be associated with reusable **Movie** and **Show** data objects to support consistency and future reuse.

US-002: Check Show Availability

User Story

The system or booking agent should be able to verify show availability in order to ensure seats are available for booking.

Implementation Details

An **Availability** stage should be introduced in the case lifecycle. Within this stage, the system or **Booking Agent** persona should verify whether the requested show has available seats. Properties such as **Seat Availability Status** and **Available Seats Count** should be captured. Booking should proceed only when seats are available, enforcing the necessary validation logic before moving forward.

US-003: Calculate Booking Cost

User Story

The system should be able to calculate the total booking cost in order to provide pricing details to the customer.

Implementation Details

Within the Availability stage, a step should be configured to calculate booking cost. Properties such as **Ticket Price** and **Number of Tickets** should be used to derive a calculated field **Total Cost** through a business rule. This calculated value should be automatically updated and stored within the case, ensuring accurate pricing details are available for review.

US-004: Confirm Booking Request

User Story

A customer should be able to confirm or cancel the booking request in order to proceed with or stop the booking.

Implementation Details

An **Approval** stage should be introduced to capture the customer's confirmation. The step should be assigned to the **Customer** persona and should capture the decision using a property such as **Booking Status**. Confirmed bookings should proceed to ticket processing; cancelled requests should be resolved appropriately without further action.

US-005: Maintain Movie and Show Data

User Story

The system should maintain movie and show details in order to enable reuse and manage show schedules effectively.

Implementation Details

Reusable data objects named **Movie** and **Show** should be defined to manage movie-related and show-related information independently from the case. These data objects should include properties such as **Movie Name**, **Genre**, **Show Date**, **Show Time**, and **Seat Capacity**. The data objects should be associated with the Movie Ticket Request case type to ensure consistency across multiple booking requests.

US-006: Review Booking Details

User Story

A customer should be able to review booking details in order to make an informed confirmation.

Implementation Details

The confirmation step should be configured to display booking information clearly to the **Customer** persona. The interface should present details such as **Movie Name**, **Show Timing**, **Number of Tickets**, and **Total Cost** in a structured format, ensuring the customer has complete visibility before confirming.

US-007: Process Ticket Booking

User Story

The system should process ticket bookings in order to finalise seat allocation and ticket generation.

Implementation Details

A **Booking Execution** stage should be added to handle final booking activities. Within this stage, the system should allocate seats and update booking status. Properties such as **Booking Confirmation Status**, **Seat Numbers**, and **Ticket ID** should be maintained within the case to ensure the booking is completed and recorded for tracking purposes.

US-008: Notify Booking Confirmation

User Story

A customer should receive a notification when the ticket booking is confirmed in order to stay informed.

Implementation Details

A correspondence mechanism should be configured to notify the **Customer** persona upon successful booking completion, triggered when the case is resolved. A **Correspondence** rule should be defined to generate the notification, including relevant details such as Case ID, Movie Name, Show Date and Time, Seat Numbers, and Total Cost.

US-009: Define Booking SLA

User Story

The system should define a service-level agreement (SLA) for each booking request so that bookings are confirmed within an expected timeframe.

Implementation Details

A simple SLA should be configured on the **Movie Ticket Request** case type. The SLA should define a **Goal** of 1 day and a **Deadline** of 2 days from case creation. When the goal is missed, the case should be flagged as approaching deadline. When the deadline is missed, the case priority should be automatically increased. No complex escalation paths are required — configuring the goal and deadline with the appropriate urgency settings is sufficient.

US-010: Route Booking Request by Show Type

User Story

The system should automatically route booking requests to the correct work queue based on the show type, so that the right team processes each booking.

Implementation Details

Within the **Booking Execution** stage, the case should be routed to one of two work queues based on the value of a **Show Type** property: **PremiumShowQueue** for premium or special screenings, and **StandardShowQueue** for all other shows. A **When** rule or a **Decision Table** may be used to implement this logic. The routing should happen automatically without any manual intervention.

Reference: Sample Correspondence (Email Body)

Subject: Movie Ticket Booking Confirmed – [Case ID]

Dear [Customer Name],

Your movie ticket booking has been successfully confirmed.

Below are the details of your booking:

- Case ID: [Case ID]
- Movie Name: [Movie Name]
- Show Date & Time: [Show Date & Time]
- Number of Tickets: [Number of Tickets]
- Seat Numbers: [Seat Numbers]
- Total Cost: [Total Cost]

Please arrive at the theatre before show time and present your booking details at entry.

Thank you for choosing our services. Enjoy your movie!

Regards,

CineWave Entertainment – Booking Support Team